

TRACE: Memory in Motion

In-Context Learning with Recurrent Memory

DATAFORGE 2026 × PATHWAY TRACK • PROBLEM STATEMENT 1 SUBMISSION • DUAL-ENGINE PARITY VERIFIED

The Central Falsifiable Claim: A fixed-size recurrent state can absorb an unbounded number of key-value demonstrations without growing in size, but when two demonstrations share overlapping (non-orthogonal) keys, the readout for one leaks the value of the other — a measurable, reproducible form of forgetting caused by fixed capacity, not a scripted failure. In our controlled experiment, increasing the state dimension allows construction of more mutually orthogonal keys and therefore reduces the interference we observe.

1. THE COMPUTATIONAL SUBSTRATE

Standard Transformer LLMs require expanding $O(T)$ KV caches to remember context. TRACE instantiates recurrent in-context learning where memory is an evolving matrix of invariant size:

- **Recurrent State:** $S \in \mathbb{R}^{d \times d}$, initialized to $S_0 = 0_{d \times d}$ (invariant d^2 scalars, independent of sequence length T).
- **Demonstration Update:** $S_t = S_{t-1} + v_t k_t^T$ (rank-1 outer product write; standard fast-weight convention).
- **Normalized Readout:** $\hat{y} = (S_t \cdot k_q) / (k_q \cdot k_q)$.
- **Derived Leakage:** $\hat{y}_a = v_a + \sum_{b \neq a} [(k_a \cdot k_b) / (k_a \cdot k_a)] v_b$. Leakage is analytically proven by linear algebra, not simulated by arbitrary noise or scripted failure.

2. KEY EXPERIMENTAL FINDINGS (FROZEN FIXTURES)

1. **Lossless Recall (d=3):** Absorbing red→1, blue→2, green→3 into $S_3 \in \mathbb{R}^{3 \times 3}$. With orthonormal keys ($k_i \cdot k_j = 0$), querying green returns **3.000** with **0.000 error**.
2. **Interference from Overlapping Keys (d=2):** Demonstrating $k_A \rightarrow 1.0$ and $k_B \rightarrow 9.0$ with $k_A \cdot k_B = 0.60$. Querying k_A yields **6.400** ($1.0 + 0.6 \times 9.0$), matching analytical leak coefficient 0.60 to machine precision.
3. **Capacity Threshold (d ∈ [2, 6]):** For $N=4$ demonstrations, MSE is substantial at $d=2$ (3.13), but drops to exact machine precision (1.26×10^{-31}) at $d \geq 4$ where orthogonal basis allocation becomes possible.

3. DUAL-ENGINE VERIFICATION & PARITY

- **Dual-Engine Parity:** Python core (core/linear_memory.py) and TypeScript twin (web/src/memory/linear_memory.ts) match to $< 10^{-9}$ tolerance.
- **Unit Tests:** 10/10 Python unit tests pass via pytest; 4/4 TypeScript contract tests pass via Node test runner.
- **Interactive Explainer:** React + TypeScript single-page app with live $d \times d$ StateGrid recomputed in $< 1\text{ms}$.
- **Reproducible Notebook:** notebook/recurrent_memory.ipynb runs top-to-bottom and reproduces golden figures.

4. LITERATURE GROUNDING & HONEST SCOPE BOUNDARY

Primary Papers (2022–2026):

- **BDH-CQ (2026):** Engdahl, Kosowski, Chorowski, Stamirowska et al., *arXiv:2608.09888*. §3.2 defines general update $S_t = U_\theta(S_{t-1}, D_t)$ and named additive special case $S_t = S_{t-1} + U_\theta(D_t)$.
- **The Dragon Hatchling (2025):** Kosowski, Uznański, Chorowski, Stamirowska, & Bartoszkiewicz, *arXiv:2509.26507*. Biological foundations of fast synaptic state.
- **Mamba (2023):** Gu & Dao, *arXiv:2312.00752*. Theoretical analysis of fixed-capacity recurrent state and compression limits.

Foundational Mechanism (Pre-2022): Schlag et al. (2021) & Katharopoulos et al. (2020) — linear attention as fast-weight associative memory.

Explicit Non-Claim: TRACE is an educational toy model instantiating the additive special case named in BDH-CQ §3.2. It does *not* reconstruct BDH-CQ's proprietary, trained U_θ function, sparsity profile (~5% active), or ARC-AGI model weights.